

Disposition of the Detailed Paper Review

Conservative Revision Dated 8 August 2026

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Scope and decision rule

The revision uses `jmlr2025_beta_mix2_04202026.tex` as its base and records accepted edits in `jmlr2025_beta_mix2_08082026.tex`. Only localized corrections to definitions, notation, grammar, typography, and plainly erroneous algebra or algorithm statements were adopted. Changes that would introduce or strengthen assumptions, rebuild a theorem, alter a rate, or replace a proof were deferred. All accepted manuscript text is shown in red.

Partially applied means that a low-risk local component was corrected, while the substantive remainder of the recommendation was deliberately left unchanged. The two proposed assumption replacements concerning quantile curvature and low-dimensional Lipschitz spatial fluctuation were not adopted.

Part I. Applied changes

ID	Issue	Disposition	What was changed and why
A. Mathematical, notation, and algorithm items			
T01	Missing data-generating model	Applied	Added the explicit equation $y_{ij} = f^*(x_{ij}) + \delta_i(x_{ij}) + e_{ij}$ and separated it from the conditional-quantile statement. This supplies a missing definition without changing the dependence assumptions.
T02	Malformed empirical inner product and domains	Applied	Defined the weighted empirical inner product at the observed x_{ij} values and changed population norms and integrals to $[0, 1]^d$ with respect to Q .
T20	Multivariate notation in the univariate trend-filtering application	Applied	Stated $d = 1$ for this application and defined $J_k(f) = \text{TV}(f^{(k-1)})$ on $[0, 1]$.
T21	Trend-filtering degree and derivative indexing	Applied	Used the consistent degree- $k - 1$ convention and corrected the piecewise-constant, linear, and quadratic examples.
T23	Duplicated trend-filtering theorem setup	Applied	Removed the repeated estimator/setup and retained one statement referring to the already defined estimator. No rate or proof was replaced.
T26	Wrong coefficient in the linear-model constraint	Partially applied	Replaced $\ \beta\ _1$ by $\ \beta^*\ _1$. The intercept and bounded-predictor questions were not resolved because they require modeling choices from the authors.
T43	Unbounded ADMM “dual update”	Applied	Replaced the positive-quadratic argmax by the standard scaled additive update $u^{(t)} = u^{(t-1)} + \theta^{(t)} - z^{(t)}$.
T44	Ambiguous check-loss proximal cases	Applied	Introduced $q_{ij}^{(t-1)}$ and γ_i and wrote the three coordinate-wise proximal cases explicitly.
T45	Errors in the K -NN ADMM statement	Applied	Corrected the dual update, the explanatory sentence, and the algorithm caption so that they refer to quantile K -NN fused lasso.
T46	Inconsistent population Huber-loss measure	Applied	Defined the population loss as an integral over $[0, 1]^d$ with respect to the design law Q .
T47	Inconsistent excess-loss substitution	Partially applied	Corrected the two displayed baselines to $\rho_\tau\{\delta_i(x_{ij}) + e_{ij}\}$. A global reconstruction of every excess-process definition was deferred.
T48	Incorrect population-minimizer wording	Applied	Replaced the minimizer claim by the direct excess-risk properties $Z(f^*) = 0$ and $Z(f) \geq 0$.
T52	Mismatched notation in the population expectation corollary	Applied	Removed the internal annotation and rewrote the short argument using the definitions of $\bar{Z}$ and Δ^2 .
T53	Literal comments and malformed localized class	Partially applied	Introduced the notation $\mathcal{F}_t = \{f \in \mathcal{F} : c_0 \Delta^2(f - f^*) \leq t^2\}$ and removed the local typeset directives in the affected lemmas. The surrounding symmetrization proof was not rewritten.
T62	One-sided localization indicators	Applied	Inserted absolute values in both occurrences of the small- and large-deviation partition.
T72	Candidate-free K -NN Rademacher supremum	Applied	Replaced θ_{ij}^* under the supremum by the centered candidate $\theta_{ij} - \theta_{ij}^*$.
T81	Even-block complexity used in an odd-block bound	Applied	Replaced $R_{\mathcal{J}_{e,l}}$ by the matching odd-block complexity $R_{\mathcal{J}_{o,l}}$.
B. Grammar, wording, and presentation items			
G01	“Broad prototyped function class”	Applied	Changed to “broad class of candidate functions” in the revised abstract.
G02	Abstract fragments for convex and nonconvex classes	Applied	Recast the fragments as complete sentences distinguishing the two settings.
G03	Unsupported promotional wording	Applied	Removed the “pioneering advancement” claim and stated only that risk bounds are derived.
G04	Unsupported real-data claim	Applied	Limited the abstract to the simulated temporal-spatial experiments actually present in the manuscript.

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ID	Issue	Disposition	What was changed and why
G05	Vague error-distribution wording	Applied	Replaced the phrase with a direct statement about applicability to normal and heavy-tailed errors.
G06	Fragmented model paragraph	Applied	Split the model, target function, and estimation goal into complete sentences.
G07	Fragmented notation paragraph	Applied	Rewrote the paragraph as complete definitions of Q , the norms, and the empirical inner product.
G08	Malformed conditional-quantile phrase	Applied	Changed it to “the conditional τ -quantile.”
G09	“Without any moments condition requirements”	Applied	Changed to “without moment assumptions.”
G10	Blocking-paragraph grammar	Applied	Replaced “Denote for” by “define” and “has the order” by “is of order.”
G11	“Release the condition”	Applied	Changed to “relax the requirement.”
G12	“Boarder class” typo	Applied	Changed to “broader class.”
G13	“The A_1 equals to”	Applied	Replaced by a complete sentence defining the role of A_1 .
G14	Article and number disagreement	Applied	Changed to “to obtain sharp convergence rates.”
G15	Repetitive proof-overview paragraph	Applied	Separated the loss comparison, spatial decomposition, symmetrization, and metric comparison into short paragraphs.
G16	Incorrect derivative ordinal and indexing	Applied	Replaced the ambiguous ordinal prose by the explicit $(k - 1)$ derivative convention.
G17	Run-on neural-network literature list	Applied	Rewrote it as a focused sentence with proper citation spacing.
G18	Ungrammatical neural-network subject	Applied	Changed to “the neural-network estimator achieves.”
G19	“Quantial” typo	Applied	Corrected the visible instances to “Quantile.”
G20	Subject–verb disagreement in the function-class setup	Applied	Changed to “We assume that this function class satisfies.”
G21	“Which signifying that”	Applied	Replaced by “meaning that” in a complete sentence.
G22	“Later case”	Applied	Changed to “latter case” and smoothed the sentence.
G24	Definition of x^k	Applied	Replaced the fragment by a complete coordinate-replacement definition.
G25	Missing space in Assumption 3	Applied	Inserted the missing space before “for all x .”
G26	“Theorem 1 replies on”	Applied	Changed both instances to “Theorem 1 relies on.”
G27	Comma splice and capitalization in the tail example	Applied	Recast the condition as “Suppose that ... hold. Then ...”.
G28	Repeated word in the tail example	Applied	Removed the repeated “condition.”
G29	“Addition term”	Applied	Changed to “additional term” and rewrote the surrounding sentence.
G30	“logn” typography	Applied	Standardized the affected expression to $\log n$.
G31	Duplicated theorem prose and “And let”	Applied	Removed the duplicate setup and used one grammatical theorem statement.
G33	“Has showed”	Applied	Changed the affected K -NN sentences to “has shown” or “shown.”
G34	Awkward plural x_{ij} s	Applied	Replaced it with “the design points x_{ij} .”
G35	“Under Assumptions 7”	Applied	Changed to the singular “Under Assumption 7.”
G36	“Gurantees” typo	Applied	Changed to “guarantees.”
G37	“Our papar” typo	Applied	Changed to “Our paper.”
G38	“An hierarchical”	Applied	Changed to “a hierarchical.”
G39	“Same to the rate”	Applied	Changed to “same as the rate.”
G40	Algorithm spacing and duplicated words	Applied	Rewrote the affected algorithm sentence and removed the duplicated “is.”
G41	Simulation-design typos	Applied	Changed to “vary n over” and “for each fixed value.”
G42	Signal-description spelling and wording	Applied	Described the function as Doppler-type with spatially inhomogeneous smoothness.
G43	Repeated panel label	Applied	Corrected the third panel reference from panel b to panel c.
G44	Misleading “i.i.d.-style pairs”	Applied	Replaced it with “flattened input–response pairs.”

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ID	Issue	Disposition	What was changed and why
G47	“Combines” at the start of proof sentences	Applied	Changed the identified occurrences to “Combining.”
G48	“Differed on” in ghost-sample prose	Applied	Changed the identified occurrences to “differ only in.”
G49	“On the both side”	Applied	Changed the identified occurrences to “on both sides.”
G50	“The rest poofs”	Applied	Changed to “The remaining proofs.”
G51	“It reminds to analyze”	Applied	Changed to “It remains to analyze.”
G52	“Seperately” typo	Applied	Changed to “separately.”
C. Experiment item			
E06	Flattening dependent observations	Partially applied	Removed the misleading “i.i.d.-style” wording. The substantive questions about mini-batches, validation splits, and dependence-aware uncertainty remain unresolved.

Part II. Suspicious or deferred changes

ID	Issue	Disposition	Reason for deferral
A. Mathematical and theoretical items			
T03	Conditional quantile definition	Deferred	Replacing exact calibration by an infimum definition changes the target and the cancellation used later in the proof. It requires an author-approved continuity/uniqueness condition.
T04	Blocking construction	Deferred	A new interlaced partition would change the coupling setup and every block-normalized proof term; this is not a local notation fix.
T05	Undefined localization radius in Theorem 1	Deferred	Introducing a critical radius would replace the main theorem and require a complete fixed-point proof.
T06	Unsupported generic polylogarithmic rate	Deferred	Removing or replacing this claim requires reconciling all application-specific entropy calculations and theorem statements.
T07	Scalar “projection” used as a function	Deferred	Selecting a near-best approximating function changes the nonconvex setup and all subsequent localization arguments.
T08	Star-hull definition	Deferred	Centering and redefining the star hull changes the class used throughout the nonconvex theorem and proof.
T09	Independence versus beta mixing	Deferred	Resolving the contradiction requires a new joint stochastic assumption, not editorial repair.
T10	Separate Berbee couplings	Deferred	Coupling the joint cluster instead would require replacing the coupling construction and checking every equality in the coupled world.
T11	Within-cluster dependence and design law	Deferred	The recommended conditional-independence and design assumptions materially strengthen the model and must be justified by the authors.
T12	Proposed local quantile-curvature assumption	Deferred	The proposed scalar uniform condition is one of the expressly rejected assumption rewrites. It changes both the hypothesis and curvature proof.
T13	Proposed low-dimensional Lipschitz fluctuation assumption	Deferred	The proposed D_i formulation and uniform block bound are not established by the current manuscript and would introduce a new substantive assumption.
T14	Missing threshold dimension in the grid proof	Deferred	The correction depends on rebuilding the disputed fluctuation assumption and its finite-grid argument.
T15	Tail constant versus curvature margin	Deferred	Fixing a quantitative absorption inequality changes assumptions, constants, and several proof lemmas.
T16	Circular tail example	Deferred	A valid sufficient example needs explicit conditioning and independence assumptions that are not presently verified.
T17	Uncentered Rademacher classes throughout	Deferred	Global centering would alter definitions and many empirical-process proofs; a one-line replacement would be unreliable.
T18	Block normalization	Deferred	Correcting weights requires tracing all $2L$ blocks through the theorem and proofs.
T19	Huber-loss to L_2 conversion	Deferred	The factor involving the bound B affects rates and proof constants beyond the local population-loss notation fixed above.
T22	Trend-filtering radius fixed at one	Deferred	Keeping a general radius changes the estimator class and the rate’s parameter dependence.
T24	Trend-filtering rate exponents	Deferred	The recommendation requires a new entropy calculation and fixed-point derivation.
T25	Minimax-optimality qualification	Deferred	The proper qualification depends on first validating the unresolved spatial remainder and trend-filtering rate.
T27	High-dimensional design-matrix definition	Deferred	Correcting the matrix and normalization changes the theorem’s complexity quantity and proof.
T28	Coefficient consistency and variable selection	Deferred	Restricting the claim to prediction or adding an identifiability condition is a substantive theorem change.

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ID	Issue	Disposition	Reason for deferral
T29	Linear-rate summary	Deferred	A corrected summary must be derived from the unresolved weighted design and block normalization.
T30	Graph total-variation scaling	Deferred	Changing the exponent must be propagated through the graph theorem and proof.
T31	Mutual versus symmetric K -NN graph	Deferred	The intended graph depends on the actual implementation; choosing “and” or “or” without the code would be speculative.
T32	K -NN norm scaling	Deferred	Normalizing the norm changes the theorem statement and all downstream rate comparisons.
T33	Undefined graph radius and polylogarithmic factor	Deferred	A valid replacement depends on the unresolved graph-TV scaling and theorem normalization.
T34	Out-of-sample extension and ties	Deferred	The tie-breaking rule and admissible K are method-design choices not documented in the source.
T35	Convexity contradiction in the nonconvex theorem	Deferred	Removing convexity requires verifying that the full star-hull proof is valid without it.
T36	Approximation-error supremum	Deferred	Replacing it by a best or near-best approximant changes the nonconvex theorem and proof.
T37	Two-sided nonconvex empirical process	Deferred	The proposed symmetric bound is a proof replacement, not a localized correction.
T38	Neural-network recursion	Deferred	A new recursion would redefine the modeled architecture and must match the approximation theorem used later.
T39	Uncontrolled neural-network class	Deferred	Clipping, sparsity, or weight bounds are substantive new assumptions.
T40	Hierarchical-composition assumptions	Deferred	Finiteness and bounded-domain conditions would strengthen the model and affect the approximation result.
T41	Architecture choice in the network theorem	Deferred	Selecting depth, width, and sparsity requires balancing approximation and estimation from scratch.
T42	Trend-filtering penalty order in the algorithm	Deferred	The discrete operator convention must be reconciled jointly with the implementation and the theory before changing it.
T49	Coupling-error probability	Deferred	The probability bookkeeping is part of the asymptotic proof and cannot be safely changed in isolation.
T50	Independence within coupled blocks	Deferred	A correction requires either a new conditional-independence assumption or a cluster-level symmetrization proof.
T51	Curvature outside the local neighborhood	Deferred	Rescaling the Huber threshold and constants depends on the disputed curvature assumption.
T54	Conditional symmetrization	Deferred	The required within-cluster independence is not established, and the alternative is a new cluster-level argument.
T55	Conditioning in Lemma 27	Deferred	Repairing the statement requires a consistent conditional-probability hierarchy across its proof and adjacent definitions.
T56	Remainder algebra	Deferred	The suggested repair rederives the conditional Knight-identity calculation and is therefore a proof change.
T57	Repeated redefinition of $\ell(f)$	Deferred	Choosing one penalty coefficient requires a verified quantitative tail-absorption condition.
T58	Incomplete proof text	Deferred	Deleting the notes without supplying the missing proof would conceal an unresolved gap.
T59	Ghost design with a shared random function	Deferred	A valid repair requires a cluster-level random-element formulation and new symmetrization.
T60	Average versus sum of absolute values	Deferred	The false inequality occurs inside the core fluctuation proof; its replacement depends on the exact process that can actually be controlled.
T61	Hoeffding centering	Deferred	Centering heterogeneous cluster terms requires an additional common-law assumption or an explicit deterministic remainder.

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ID	Issue	Disposition	Reason for deferral
T63	Tail-example conditioning	Deferred	Separating outer and inner probability layers requires assumptions not stated in the source.
T64	A_3 ghost-process argument	Deferred	The suggested deterministic cluster-level symmetrization replaces a substantive proof step.
T65	Circular aggregation	Deferred	Fixing the block indices, coupling remainder, and theorem dependence requires rebuilding the aggregation proof.
T66	Radius t versus t^2	Deferred	Correcting dimensions propagates through local balls, fixed points, and final rates.
T67	Malformed O_P criterion	Deferred	A valid sub-root peeling lemma and high-probability statement would replace the current argument.
T68	Trend-filtering population-to-empirical localization	Deferred	The required norm-equivalence or uniform entropy event is not proved in the source.
T69	Fixed-point algebra	Deferred	The correction requires recomputing the entropy integral and solving the scalar inequality.
T70	Trend-filtering lemma/theorem rate mismatch	Deferred	Selecting one rate without a verified derivation would be arbitrary.
T71	Odd-block denominator in the linear proof	Deferred	The denominator is entangled with unresolved block-normalization conventions.
T73	Circular local-radius assumption in the K -NN proof	Deferred	The unknown radius must be retained and a new fixed-point inequality solved.
T74	Inconsistent logarithmic exponents and symbols	Deferred	A single exponent and K choice must follow from a checked graph complexity proof.
T75	Broken K -NN extension proof	Deferred	The unresolved reference, swapped terms, and radii require a full reorganization of the proof.
T76	Dependent-design K -NN geometry	Deferred	The needed occupancy and degree proposition under mixing is not available in the current source.
T77	Graph lemma equality and norm errors	Deferred	Fixing the pseudoinverse and norm conventions changes multiple graph lemmas and their uses.
T78	Nonconvex localization defects	Deferred	A near-minimizer and corrected two-sided processes require rebuilding the nonconvex proof.
T79	Inconsistent approximation-adjusted radius	Deferred	A single definition cannot be selected safely until the surrounding nonconvex proof is red-erived.
T80	Sign errors in the twelve-term decomposition	Deferred	Replacing the decomposition by an absolute centered process is a major proof rewrite.
T82	Missing weights and inconsistent expectations	Deferred	The correction must be propagated through all nonconvex process terms and penalty coefficients.
T83	Neural-network entropy/application proof	Deferred	The explicit architecture, approximation theorem, entropy bound, and fixed point all remain to be established together.
T84	Berbee coupling proof	Deferred	Quoting or proving sequential joint coupling changes a foundational proof and must be checked against the precise mixing assumptions.
T85	Bibliographic accuracy	Deferred	Metadata and cited theorem locations must be verified against primary sources; guessing would be unsafe.
B. Grammar, wording, and presentation items			
G23	Repeated “any” and proposed scalar perturbation	Deferred	The duplicate word is absent from the base source. The remaining recommendation would rewrite the disputed curvature assumption.
G32	Alleged missing spacing in the linear-theorem discussion	Deferred	The cited string does not occur in the base source, so there is no supported local edit to make.
G45	Acknowledgments template text	Deferred	Verified funding and competing-interest language was not supplied.
G46	Internal directives in the proof appendix	Deferred	Broad deletion would hide unresolved mathematical gaps. Only comments attached to the locally corrected notation in the applied section were removed.

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ID	Issue	Disposition	Reason for deferral
G53	Global capitalization	Deferred	A manuscript-wide style pass is broader than the requested conservative corrections and risks changing technical naming inconsistently.
G54	Global temporal-spatial terminology	Deferred	Choosing one term throughout is an authorial style decision and would create a large, non-local diff.
G55	Global hyphenation and spacing cleanup	Deferred	This requires a separate full copyedit after the mathematical text is settled.
C. Experiment-specific items			
E01	Inconsistent basis-function definition	Deferred	The scalar basis actually used must be confirmed from the experiment code.
E02	Unequal- m_i design recursion	Deferred	A padded-array construction would change the simulation design and must match the implementation.
E03	Quantile target under correlated noise	Deferred	The centering or quantile shift used in each scenario is not documented and cannot be inferred safely.
E04	BIC sample size and weights	Deferred	The effective sample size and weighted criterion must be confirmed from the code.
E05	Degrees-of-freedom threshold and notation	Deferred	The threshold needs a source or sensitivity analysis, and the intended knot count must be confirmed.
E07	Missing Scenario 5	Deferred	It is unclear whether this is a numbering error or an omitted experiment.
E08	Scenario 8 quantile target	Deferred	The oracle quantile functions for the evaluation grid are not supplied.
E09	Monte Carlo uncertainty	Deferred	Standard errors and replication counts require the experiment outputs or rerunning the simulations.
E10	Neural-network reproducibility	Deferred	Architecture, optimization, seeds, and tuning budgets must come from the implemented experiment.
E11	Squared-error baseline interpretation	Deferred	Adding competitive baselines changes the experiment and requires new computation.
E12	Funding and conflict disclosure	Deferred	Only the authors can provide verified disclosure language.

Conclusion

The applied set is intentionally narrow. Items in the second part should remain open until the authors choose the intended stochastic model and verify the affected theorem and proof chains. In particular, no new curvature or low-dimensional fluctuation assumption has been inserted into the manuscript.